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Git Submodules

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What Are Git Submodules?

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Git submodules let you include one Git repository inside another as a subdirectory.

This is useful for adding libraries or dependencies managed in separate repositories, while keeping their commit history separate.

Why Use Submodules?

Submodules are helpful when you want to:

- Reuse code from another project
- Track a library or dependency at a specific commit
- Keep histories of projects separate

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Example: Add a Submodule

```
[user@localhost] $ git submodule add https://github.com/example/library.gi
```

This creates a subdirectory `libs/library` and updates `.gitmodules` with the submodule info.

How to Clone a Repo with Submodules

When you clone a repository with submodules, you need to fetch their contents separately:

Example: Init and Update Submodules

```
[user@localhost] $ git submodule init
[user@localhost] $ git submodule update
```

Or do it all at once when cloning:

Example: Clone with Submodules

```
[user@localhost] $ git clone --recurse-submodules https://github.com/user/
```

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How to Check Submodule Status

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```
[user@localhost] $ git submodule status
```

How to Run Commands in All Submodules

You can run a command in every submodule. For example, to check their status:

Example: foreach

```
[user@localhost] $ git submodule foreach git status
```

How to Update Submodules

To update submodules to the latest commit from their remote repository:

Example: Update All Submodules

```
[user@localhost] $ git submodule update --remote
```

How to Remove a Submodule

To remove a submodule:

- Delete the relevant section from `.gitmodules`
- Remove the submodule directory from your working tree
- Run `git rm --cached path/to/submodule`

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or remove submodules.

Example: .gitmodules File

```
[submodule "libs/library"]
  path = libs/library
  url = https://github.com/example/library.git
```

Troubleshooting and Best Practices

- If submodules are empty after cloning, run `git submodule update --init --recursive`.
- If you change a submodule's URL, update both `.gitmodules` and `.git/config`.
- Submodules always point to a specific commit, not always the latest-remember to update if you want new changes.
- Keep submodules for external projects you want to track at a fixed version. For simpler needs, consider alternatives like Git subtree or copying files.

Note: Submodules are powerful, but can be tricky to manage.

Only use them if you really need to track another project at a specific commit.

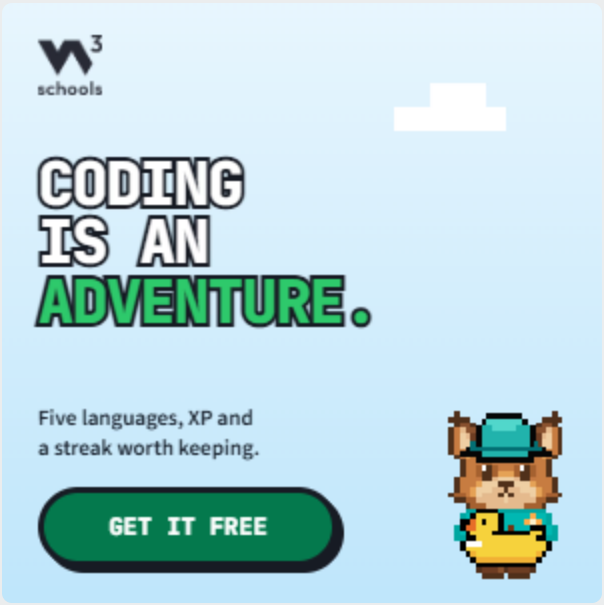
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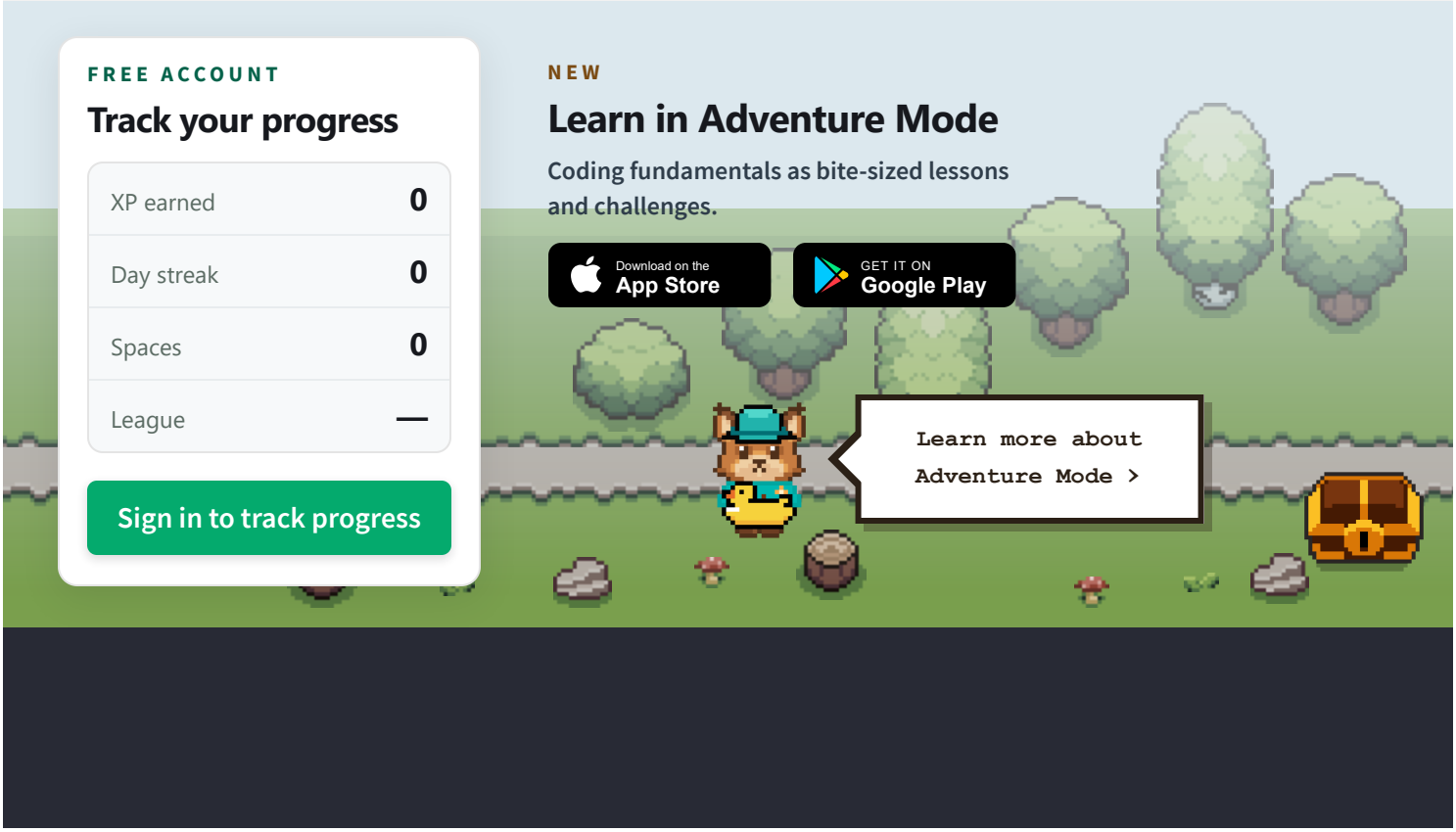
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